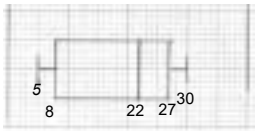
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	10 : 3 final answer	**2**	**M1** for 1500 : 450 oe in ratio form If 0 scored **SC1** for answer 3 : 10
1(b)	360 240 400	**3**	**B2** for answer 0.36 0.24 0.4 or for answer two of 360 240 400 or **M1** for $\frac{1000}{9+6+10}[\times k]$ where k = 1, 9, 6 or 10 If 0 scored, **SC1** for answer with 3 values in ratio 9 : 6 : 10 in that order
1(c)	3.68 cao	**2**	**M1** for $\left(1+\frac{15}{100}\right)\times 3.2$ oe or **B1** for answer 0.48
1(d)	18 804[.0...]	**2**	**1** for $16\,620\times\left(1+\frac{2.5}{100}\right)^5$ oe
1(e)	3.95	**3**	**M2** for 22.5 – (18.5 to 18.6) or (22 to 23) −18.55 or **M1** for 23 – 0.5 oe seen or 23 + 0.5 oe seen or 18.5– 0.05 oe seen or 18.5 + 0.05 oe seen
2(a)	142 142	**2**	**B1** for each **FT** angle *b* = *their* angle *a*
2(b)	150	**2**	**M1** for $\frac{360}{12}$ oe isw or $180\times(12-2)$ oe isw

Question	Answer	Marks	Partial Marks
2(c)	56	**B1**	
	34	**B2**	**M1** for angle at centre = 2 × *their* 56 oe soi or for angle *OMB* = 90 oe soi
2(d)	51	**2**	**B1** for opp angle = 129 soi
3(a)(i)	25.4375	**4**	**M1** for mid-points soi (5, 17.5, 32.5, 50) **M1** for use of Σ*fm* with *m* in correct interval including both boundaries **M1** for (dep on 2nd M1) for Σ*fm* ÷ 40
3(a)(ii)	correct histogram	**3**	**B2** for 3 correct blocks or **B1** for 2 correct blocks If 0 scored **SC1** for 4 correct frequency densities 0.3, 1.2, 1, 0.2 oe soi
3(a)(iii)	$\frac{19}{260}$ oe	**3**	**M2** for $\frac{19}{40} \times \frac{3}{39}[\times 2]$ oe or **M1** for any of $\frac{19}{40}$, $\frac{3}{40}$, $\frac{19}{39}$, $\frac{3}{39}$ oe seen If 0 scored, **SC1** for $\frac{57}{800}$ oe
3(b)(i)	5	**1**	
3(b)(ii)	 Correct box plot	**3**	**B2** for with LQ at 8 and median at 22 and UQ at 27 and boxed or **M1** for LQ at 8 and median at 22 or for UQ at 27 **B1** for lowest = 5 and highest = 30 Max **B1** if not box and whisker diagram

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